

ECEN 689: Optical Interconnects

Homework #2

Due: 2-5-2026, 11:59PM

Homeworks will not be received after due.

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1. **Multi-Mode Fiber Channel.** A short distance 10Gb/s interconnect system uses a graded-index multi-mode fiber (GRIN-MMF) channel with $n_{core}=1.48$ and $n_{clad}=1.46$. What is the maximum transmission distance such that the pulse spreading due to modal dispersion is 10% of the bit period? Refer to Lecture 2 for the GRIN-MMF modal dispersion model.

2. **Single-Mode Fiber Channel.** A 10Gb/s system operating at $\lambda=1310\text{nm}$ utilizes a single-mode fiber channel with loss of 0.4dB/km and $D=0.5\text{ps}/(\text{nm}\cdot\text{km})$. The transmitter laser source has a 3nm linewidth and outputs 1mW average power. Assuming a receiver sensitivity $\bar{P}_{sens} = -15\text{dBm}$, answer the following.
 - a. What is the maximum transmission distance? Is the link loss- or (chromatic) dispersion-limited? For chromatic dispersion, assume that ΔT should be at most half the bit period.
 - b. If the fiber length is 5km, what is the maximum data rate?

3. **Vertical p-i-n Detector.** An InGaAs vertical p-i-n detector has a $2\mu\text{m}$ intrinsic region with an absorption coefficient $\alpha=10^4\text{cm}^{-1}$. The device is biased to yield carrier velocities of 10^5m/s and electrical parasitics of $R_{PD}=20\Omega$ and $C_{PD}=150\text{fF}$.
 - a. Assuming no reflection losses, what is the responsivity at $\lambda=1310\text{nm}$?
 - b. What is the total PD bandwidth, including both transit-time and RC effects?

4. **Waveguide p-i-n Detector.** A Ge waveguide p-i-n detector has a 320nm intrinsic with an absorption coefficient $\alpha=10^3\text{cm}^{-1}$ and a $10\mu\text{m}$ absorption length. The device is biased to yield carrier velocities of 10^5m/s and electrical parasitics of $R_{PD}=50\Omega$ and $C_{PD}=7\text{fF}$.
 - a. Assuming no reflection losses, what is the responsivity at $\lambda=1310\text{nm}$?
 - b. What is the total PD bandwidth, including both transit-time and RC effects?

5. **Simple Resistive Front-End Sensitivity.** A simple front-end with noise bandwidth $BW_n=10\text{GHz}$ is constructed with a 50Ω resistor. This front-end is used with 3 effective photodetector configurations.
 - a. p-i-n detector with $R=1\text{A/W}$
 - b. APD with $R=1\text{A/W}$, $M=8$, $F=4$
 - c. OA + p-i-n with $R=1\text{A/W}$, $G=50$, $\eta F=2$
 For these 3 effective photodetector configurations:
 - i. Give the optical sensitivity for a $\text{BER}=10^{-12}$ considering both amplifier and detector noise. Hint: You need to first compute the “amplifier” noise, which is the resistor rms current noise over a 10GHz bandwidth. Assume $T=300\text{K}$.
 - ii. With the computed sensitivity and assuming a high extinction ratio, what is the high-level detector shot noise, $i_{n,X,1}^{rms}$?